2018A F=ma Exam: Problem 20

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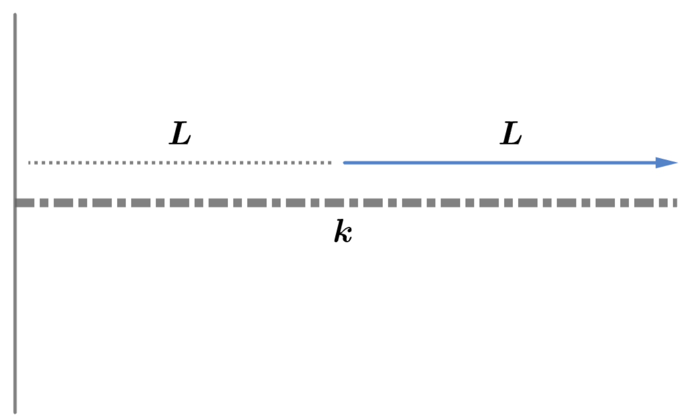

Recall the potential energy stored in a spring is given by

$$U = \frac{1}{2}kx^2$$

so we have

$$U_0 = \frac{1}{2}kL^2$$

A spring consists of two 'half-springs' in series so

$$\frac{1}{k} = \frac{1}{k_{1/2}} + \frac{1}{k_{1/2}}$$

Thus, the spring constant of a 'half-spring' is $k_{1/2} = 2k$. If two 'half-springs' are stretched to double their original length, the total potential energy is

$$U = 2 \times \frac{1}{2}(2k)\left(\frac{L}{2}\right)^2 = U_0$$

so the answer is $\boxed{\text{C}}$.